Le Chiffre

Write a Python program that implements a generic Vigenère cipher that takes a ciphertext as input from `stdin` and outputs the correct plaintext and corresponding key used in the cipher to `stdout`. You will be provided with a dictionary of words that is **guaranteed** to contain the key used to generate the ciphertext. Note that the key used to generate the ciphertext can include any of the characters in the alphabet (e.g., punctuation).

Since you will be playing the part of cryptanalysts, your program will need to generate candidate plaintexts using a provided alphabet and keys in the provided dictionary. Note that the alphabet can be hard-coded in your program. Once candidate plaintexts are generated, you must **efficiently** filter out invalid ones using a method of your choice. As in the previous program, one way is to compare “words" in candidate plaintexts with words in the provided dictionary. Note that you may want to “normalize" words in candidate plaintexts and the dictionary (e.g., remove punctuation, convert to lowercase, etc). If enough words match (say, three-fourths of them), a candidate plaintext becomes likely. You can also consider using the average length of words in English US to help identify candidate plaintexts.

Notes and Requirements:

- Submit your source code only. I will provide my own ciphertexts to test with;
- Read the ciphertext from stdin;
- Write the plaintext and shift to stdout;
- Comment your source code appropriately;
- If the ciphertext contains characters not in the alphabet (e.g., \n), leave them as is in the candidate plaintexts; and
- The ciphertext could contain multiple lines.

Please, no GUIs. Make this a command line application without frills that I can execute at the command line as illustrated below via several sample runs of my program:

```
mtahat>py le-chiffre.py < ciphertext-1.txt
KEY=Byzantine
The lady said, "Oh my! You have nice eyes. Are they yours?"
I laughed.

mtahat>py le-chiffre.py < ciphertext-2.txt
KEY=Cthulhu's
Never attribute to malice that which is adequately explained by stupidity.
(Hanlon's razor)
```